

SpendSense AI: Financial Leakage Detection for Corporate Professionals

Executive Business & Technical Analysis Report

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1. Executive Summary & Business Problem

Salaried corporate professionals in India (earning between ₹35,000 and ₹1,50,000 per month) frequently struggle with 'saving leakages'—small, frequent, non-essential expenditures that accumulate over time and silently deplete monthly savings. While large fixed expenses (like rent, utilities, and EMIs) are easy to track, discretionary habits such as weekend dining, frictionless UPI/Credit Card transactions, duplicate subscriptions, and office-commute coffee runs go unnoticed.

SpendSense AI is an end-to-end data analytics and machine learning solution designed to address this problem. By analyzing a simulated dataset of 500 corporate employees over 12 months (~156,700 transactions), this project: identifies where unnecessary money is being spent, isolates spending behavior based on work-mode and salary bands, trains a machine learning model to predict potential expense leakage, and offers targeted recommendations to help users improve savings.

2. Dataset & Methodology

Disclaimer: The dataset used is synthetically generated for educational and portfolio demonstration purposes.

The dataset models a diverse corporate workforce categorized into four Salary Bands:

- **Band A (₹35k-50k):** Entry-level Analyst/Associate (~40% weights)
- **Band B (₹50k-80k):** Senior Analyst/Specialist (~30% weights)
- **Band C (₹80k-120k):** Team Lead/Consultant (~20% weights)
- **Band D (₹120k-150k):** Manager/Senior Consultant (~10% weights)

The simulation models rent, utilities, EMIs, and monthly SIP investments, combined with discretionary transactions. Derived attributes like **Weekend_Flag** and **Salary_Week** (first 7 days of the month) were engineered in Python.

3. Exploratory Data Analysis (EDA) Findings

Key behaviors discovered during Python EDA include:

- **Expense Allocation:** Discretionary 'Want' spending accounts for **35.4%** of the workforce's overall spending.
- **Savings Rate:** The organizational savings rate is stable between **20.1% and 21.0%**.
- **Frictionless Spikes:** Credit Cards and UPI account for over **70%** of all transactions. Credit Cards show an average transaction value of **₹3,100** (indicating high discretionary spending).
- **Work Mode Impact:** WFH employees show the highest average daily discretionary spend (**₹2,047.45**) due to online shopping and food deliveries, compared to Office (**₹1,706.08**) and Hybrid employees (**₹1,908.80**).
- **Salary Week Splurge:** Discretionary want spending values increase by **30-50%** during the first week of the month.

4. SQL Business Analytics Insights

Running the business analysis queries yielded clear operational metrics:

- **Top Leakage Categories:** Gym memberships (**₹12.99M**), Shopping (**₹10.17M**), Dining Out (**₹9.43M**), Food Delivery (**₹4.72M**), and OTT Subscriptions (**₹2.93M**).
- **Payment Mode Risk Index:** Credit Cards carry the highest leakage rate (**41.87%** of transactions are leakages), followed by UPI (**26.71%**). Cash shows **8.67%** leakage, while Debit Cards show under **2%**.
- **Unused Subscription Bleed:** Unused annual gym subscriptions (Cult.fit, Gold Gym) and OTT platforms (Netflix, Spotify, YouTube Premium) accounted for over **₹15.9M** in leakage.

5. Machine Learning Classifier Performance

To build a technically correct classification pipeline, the leakage target was created using rule-based metrics, but the features leakage_score and Expense_Type were dropped before training to prevent target leakage.

Metric	Logistic Regression	Random Forest Classifier
Accuracy	99.50%	99.85%
Precision	98.19%	99.20%
Recall	99.19%	100.00%
F1 Score	98.69%	99.60%
ROC-AUC	99.98%	100.00%

6. Actionable Recommendations & Business Impact

Salaried professionals can reclaim ****₹5,000 – ₹15,000 per month**** by adopting these targeted steps:

1. **Digital Payment Caps:** Set a monthly transaction cap of ■10,000 on credit cards for discretionary wants, and disable UPI auto-pay features.
2. **Quarterly Subscription Audits:** Conduct a regular check to identify and cancel duplicate streaming services and unused gym subscriptions.
3. **Automate Savings First:** Set up an automated Mutual Fund SIP that debits 20-30% of income on the 1st or 2nd day of the month.
4. **48-Hour Shopping Rule:** Keep items in online carts for 48 hours before purchasing to eliminate emotional impulse buying.

Impact: Cutting the top 10% of leakage expenses in half boosts the average employee's savings rate from **20% to over 26%**, leading to an additional ■60,000 – ■1,80,000 in yearly wealth accumulation.